ecoSignalLab User Guide

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`ecoSignalLab` (`esl`) is for answering a practical question: *what happened in this audio, when did it happen, and how sure are we?* It is a command-line toolkit for environmental audio, room and impulse-response work, long-duration recordings, multichannel material, calibration-aware measurements, and ML-ready exports.

This is the friendly manual. It gets you from one audio file to useful answers without requiring an acoustics degree, while leaving the rigorous details one click away when you need them.

1. The Smallest Useful Start

Install and inspect your environment:

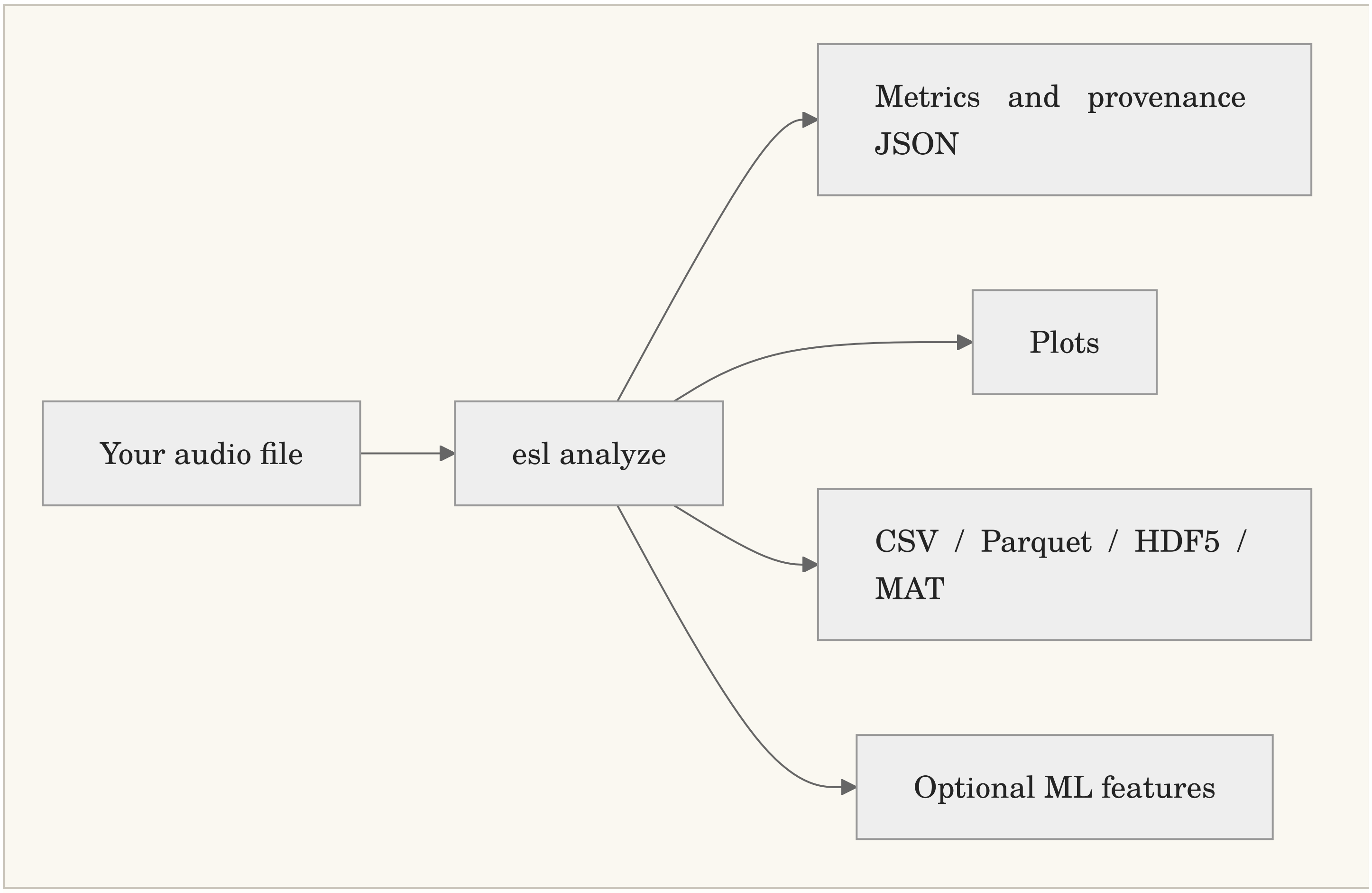
```
python3 -m pip install --user pipx
pipx install "ecosignallab[io,plot,features]"
esl doctor
```

Analyze a file and make plots:

```
esl analyze input.wav --out-dir out --json out/input.json --plot
```

This creates a JSON record of results and provenance, plus plots under `out/input_plots/`. If the command says `esl: command not found`, use the module form while you finish installation:

```
python -m esl --help
```



Plain English: one command can give you a readable report now and structured data later. You do not need to decide your entire research program before looking at the first graph.

2. What This Is and Is Not

The short version

`esl` is a command-line toolkit for making a durable, inspectable statement about an audio recording. It can answer questions such as: *when did the recording change? which interval is worth listening to? how does this room response decay? which files resemble this example? what did the recorder, decoder, calibration profile, and analysis configuration contribute to this number?* It is designed to return both a convenient answer and the context needed to challenge that answer later.

It is not a black box that converts a WAV file into truth. Audio analysis is a chain of choices: microphone, placement, gain, format, decoder, channel order, window, feature, normalization, threshold, aggregation, and interpretation. `esl` makes those choices explicit wherever it can. That is the point. A smaller result with stated assumptions is more useful than a dramatic result with no path back to the audio.

What `esl` is

A reproducible measurement workbench

Use `esl` when you need an analysis result that can travel with its method. `esl analyze` records resolved configuration, selected metrics, library versions, decoder provenance, channel information, validity flags, calibration state, and a `pipeline_hash`. The JSON is not merely an export format. It is a compact measurement record that tells a collaborator what was actually run.

For a simple file, that may feel excessive. For a set of recordings collected over months, a room-simulation comparison, or an ML training corpus, it is the difference between a result that can be checked and one that can only be remembered. The first useful habit is therefore simple: keep the command, configuration, output JSON, and source identifier together.

A multichannel-first audio analysis toolkit

`esl` accepts one channel through larger arrays without silently treating every recording as mono. It preserves per-channel results, labels aggregate values, and exposes the downmix or pooling policy used by a metric. This matters when a bird call is visible only on one microphone, when an array has a failed channel, when left and right differ, or when an architectural render carries separate receiver signals.

For Ambisonics and spatial material, `esl` can use declared metadata such as channel order, normalization, and labels. It can also use a sidecar when a filename is not enough. It does not infer a spatial convention from optimism. If the recording says four channels but does not say whether they are ACN/SN3D, FuMa, WXYZ, or something else, the correct response is to retain the uncertainty until you supply defensible metadata.

A tool for finding, reviewing, and exporting moments

Long recordings are usually not hard because they contain too little sound. They are hard because they contain too much unreviewed sound. `esl moments extract` converts a ranking policy into timestamped candidate clips and a CSV review record. You can request one candidate with `--single`, a shortlist with `--top-k`, or all detected candidates. You can set pre-event and post-event context so the exported WAV does not begin in the middle of the thing you need to understand.

Novelty, level, spectral change, or an anomaly score can prioritize review. A score is not a label. The highest-ranked clip may be a bird phrase, a door, wind on a microphone, a gain change, or the beginning of a useful question. `esl` helps make that review bounded and reproducible; it does not remove the need to listen, inspect a plot, or record a human decision.

A bridge between acoustic domains

The same recording can matter to different people for different reasons. An ecologist may need a band-energy contrast and a time series. An acoustic consultant may need an impulse-response decay estimate, clarity values, and variant comparison. An engineer may need clipping, level, spectral features, and a CSV suitable for a measurement workflow. An ML practitioner may need a FrameTable, a tensor layout, and a dataset manifest that does not leak adjacent audio between train and evaluation splits.

`esl` provides a common analysis substrate for those workflows while keeping their assumptions visible. A metric does not become ecological merely because it was calculated outdoors, architectural merely because the file was called `room.wav`, or ML-ready merely because it was saved as an array. The surrounding metadata and validation practice determine what the value can support.

A long-archive and batch-processing tool

`esl` is designed to stream, chunk, shard, and summarize audio that cannot be comfortably loaded as one giant array. A multi-day RF64 file, a directory of daily recordings, or a calendar-aware shard manifest can be processed as an ordered archive. This lets you find candidate moments, make trends, retrieve similar intervals, and write appendable feature data without pretending that years of audio belong in RAM.

Large-file support is not a promise that every operation is cheap. A full self-similarity matrix grows rapidly with the number of frames, and dense feature output can be larger than expected. `esl` exposes chunk and shard controls so you can choose an appropriate time scale and storage strategy. Read RF64 and Large Files before starting an analysis whose duration is measured in days, years, or alarming quantities of GB.

A transparent, inspectable implementation

An audio-analysis tool earns trust by showing its work. For that reason, `esl` tries to put the ordinary but decisive details in the result: which decoder opened the file, which channel count and sample rate it observed, which metric versions ran, whether calibration was applied, which validity flags were raised, and which resolved settings formed the `pipeline_hash`. A result should remain intelligible after the terminal scrollback is gone and after the person who ran the command has moved on to another project.

That does not mean every number is equally certain. A clipped signal can still have a peak value, but the clipping flag matters. An SNR estimate can still be useful, but it should travel with its confidence and method. An impulse-response fit can still report RT60, but its fit quality and detection state tell you whether that value deserves weight. `esl` is designed to preserve these distinctions in the data model rather than hiding them behind a single, overconfident summary score.

Open source is part of this contract, not a decorative label. You can inspect metric implementations, tests, formulas, attribution, and cited methods in the repository. You can also decide that a metric's assumptions do not match your question and select another one. The toolkit is intended to make disagreement productive: compare configurations, rerun a fixture, examine a plot, and say precisely where two workflows differ.

A command-line tool with optional visual evidence

The command line is the primary interface because a command can be copied into a lab notebook, a shell script, a batch job, or a methods section. It also means `esl` does not require an account, a browser tab, a remote control plane, or an always-running graphical application before you can inspect one WAV. Commands produce files that remain yours: JSON, CSV, Parquet, HDF5, MAT, WAV clips, and static or interactive plots.

That command-line choice is not a demand that users stare at text all day. Plots are evidence, not an afterthought. Spectrograms, level trends, novelty curves, similarity matrices, decay curves, and batch exports support the question that comes after a number: *does this claim look plausible in the audio itself?* Run with plotting options, open the generated artifact, and listen to the corresponding interval. A plot can expose a bad channel, a decoder surprise, a gain step, or a meaningless threshold much faster than a column of decimals.

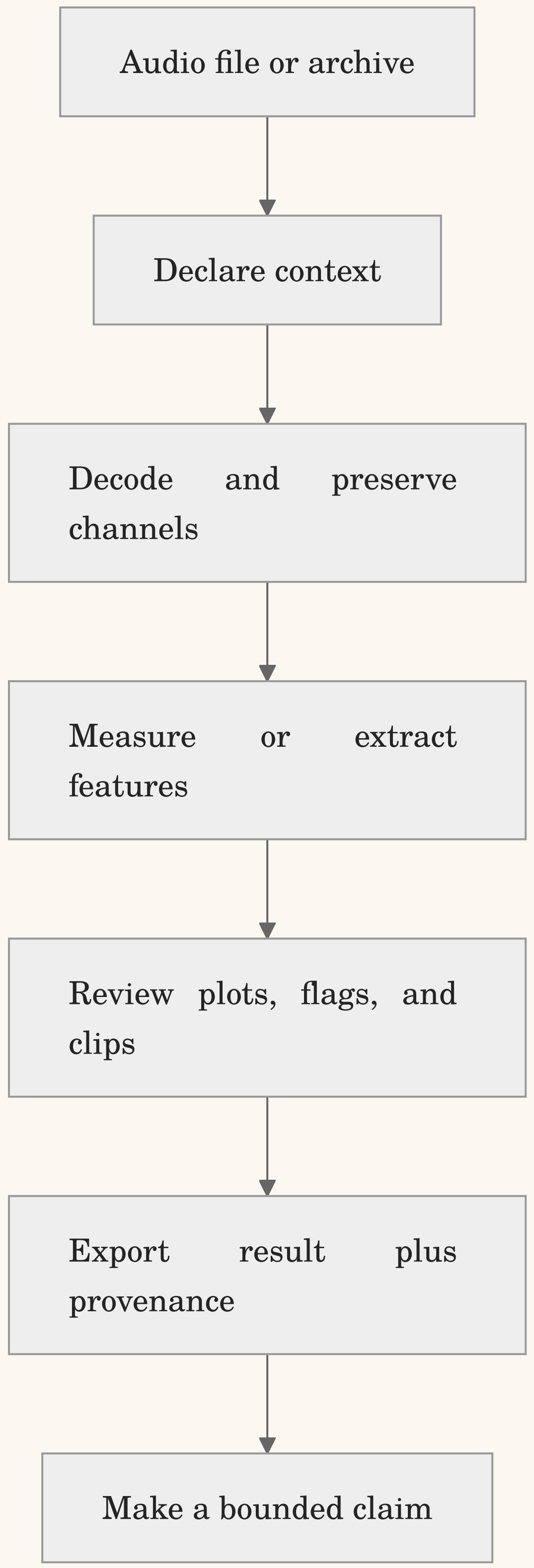
The interface therefore favors a simple progression. Type one command for a first answer. Add an output directory when you need durable artifacts. Add a calibration file, channel metadata sidecar, shards, or ML export only when the question demands them. The beginner path and the research path should share a method, not require two unrelated products.

A composable SDK, not a locked analysis service

`esl` is meant to work inside a wider practice. A field team can use it after a deployment. A consultant can use it after a room simulator exports receiver signals. A researcher can call its Python APIs from a notebook. A data engineer can build a scheduled batch workflow around its manifests and appendable FrameTables. The same output contract is intended to support all of those uses, so a quick command-line inspection does not paint a project into a corner when it grows.

This also means the toolkit should be conservative about ownership. It does not claim your recordings, prescribe a cloud platform, or insist that every workflow use its database. It aims to interoperate through documented files, schemas, and explicit metadata. If another tool is better at annotation, curation, statistical modeling, listening, or reporting, use that tool and retain the `esl` provenance beside the handoff. A healthy pipeline is usually a chain of specialized tools, not one application pretending to be all of them.

The practical payoff is freedom to begin with a single file and still retain a route to batch processing, peer review, or model training. The discipline is the same at every scale: preserve the source identity, declare the configuration, retain the output, and state what the result does and does not mean.



Plain English: `esl` is built to move from an audio file to a reviewable analysis package, not from an audio file to an unexplained verdict.

What `esl` is not

Not an audio editor or a digital audio workstation

`esl` analyzes, exports, and extracts clips. It is not intended to replace a DAW, waveform editor, non-linear video editor, or live performance tool. Use a dedicated editor when your primary job is arranging, cutting by hand, mixing, restoring a performance, or producing a finished soundtrack. Use `esl` when your primary job is measuring, ranking, comparing, documenting, or preparing audio for a reproducible downstream workflow.

This boundary is useful rather than limiting. A DAW is excellent at helping a person shape sound. `esl` is excellent at helping a person state what an analysis did to sound. You can use both: export a candidate clip with `esl`, review it in an editor, then place the human annotation back beside the CSV and provenance record.

Not a replacement for field instrumentation or a measurement protocol

`esl` can report dBFS from decoded samples without a physical calibration. SPL, dBA, dBC, and pressure-chain results require a documented calibration model. Entering `0 dBFS = 94 dBA` is an assumption supplied by a user; it is not a property discovered inside an arbitrary recording. Microphone sensitivity, preamp gain, ADC range, weighting, reference tone, recorder gain state, and deployment practice all matter when a result will be interpreted as a physical level.

Use a calibrated sound level meter, calibrator, microphone documentation, and a field or laboratory procedure when the question requires them. `esl` can carry the resulting assumptions and help verify a calibration tone workflow. It cannot repair an unknown microphone position, recover a gain setting that was never recorded, or turn clipped audio into a compliant measurement.

Not a universal sound recognizer

`esl` can produce features, similarity rankings, novelty curves, anomaly scores, and model-ready tensors. None of those automatically names a source. To recognize species, machines, instruments, spoken words, or acoustic events, you need an appropriate model, representative labeled data, a stated evaluation split, and a validation protocol that matches the deployment.

Similarity is especially easy to overread. A query result means that one file is close to another under the selected representation and distance function. It does not mean the recordings share a source, cause, location, or ecological meaning. Use the ranking to reduce review time, then listen and inspect the evidence. See [Similarity Search] and [ML Features] for the contracts behind those outputs.

Not a standards-certification engine

Some `esl` metrics are implementation-aligned measurements, some are research features, and some are explicitly labeled proxies. A proxy can be useful for comparison, screening, and hypothesis generation without being a certified substitute for a regulated procedure. RT60, clarity, loudness, and soundscape metrics all depend on signal conditions, definitions, bandwidth, fit range, calibration, and standards context.

If a project requires compliance, legal defensibility, occupational exposure assessment, building-code sign-off, or a standards-specific procedure, identify the governing standard and use a complete validated workflow. `esl` can be one component of that workflow when its output, inputs, and limitations match the requirement. It should not be presented as a certificate generator because it printed a number with three decimal places.

Not a substitute for human interpretation

Plots can reveal structure, and metrics can make comparisons repeatable, but they do not know your study question. A novelty peak does not know whether it is scientifically important. An NDSI value does not know whether a local band choice makes ecological sense. A room metric does not know which listener, source, or use case matters. A model score does not know whether its labels were biased or its evaluation split leaked time-adjacent audio.

Use `esl` to make the numerical and procedural part of a decision explicit. Then bring listening, domain knowledge, field notes, drawings, annotations, ground truth, and skeptical review to the interpretation. The right outcome is often a narrower claim, a better next measurement, or an honest statement that the current audio cannot answer the question alone.

A practical decision guide

Choose `esl` when you can complete the sentence: "I have this audio, I want to measure or compare this property, and I am willing to retain the assumptions that make the result interpretable." Start with `esl simple` or `esl analyze` for one file; use `esl moments extract` for reviewable candidate clips; use `esl similar` for retrieval; use `esl shard` for ordered archives; and use the FrameTable export when the next stage is ML.

Pause before using `esl` alone when the sentence is instead: "I need a legal certificate," "I need to know the species without labels," "I need to repair a bad recording," "I need to mix this album," or "I need a physical measurement but know nothing about the recording chain." Those are different problems. They may still use audio analysis, but they require other instruments, tools, evidence, or people.

When a result depends on calibration, decoder choice, channel convention, or a proxy assumption, `esl` records that context in the output JSON. See Schema for the exact contract and Metrics Reference for what each metric declares about units, aggregation, calibration, and streamability.

3. Five Things You Can Do Today

Analyze one recording

```
esl simple forest_morning.wav
esl analyze forest_morning.wav --out-dir out --plot --ml-export
```

Use `simple` when you want a short answer. Use `analyze` when you want durable JSON, plots, and exports.

Find the single most interesting moment

```
esl moments extract forest_morning.wav \
  --out out/moments --single --rank-metric novelty_curve --event-window 12
```

The output includes a clipped WAV and a CSV row with its timestamp. Change `--event-window`, or use `--window-before` and `--window-after`, to decide how much context travels with the event.

Find several unusual moments

```
esl moments extract forest_morning.wav \
  --out out/top_moments --top-k 20 --rank-metric novelty_curve \
  --window-before 5 --window-after 15
```

Find files that sound like a query

```
esl similar query_frog_call.wav recordings/ \
  --out out/similar --top-k 10 --feature-set all
```

Make similarity and novelty plots

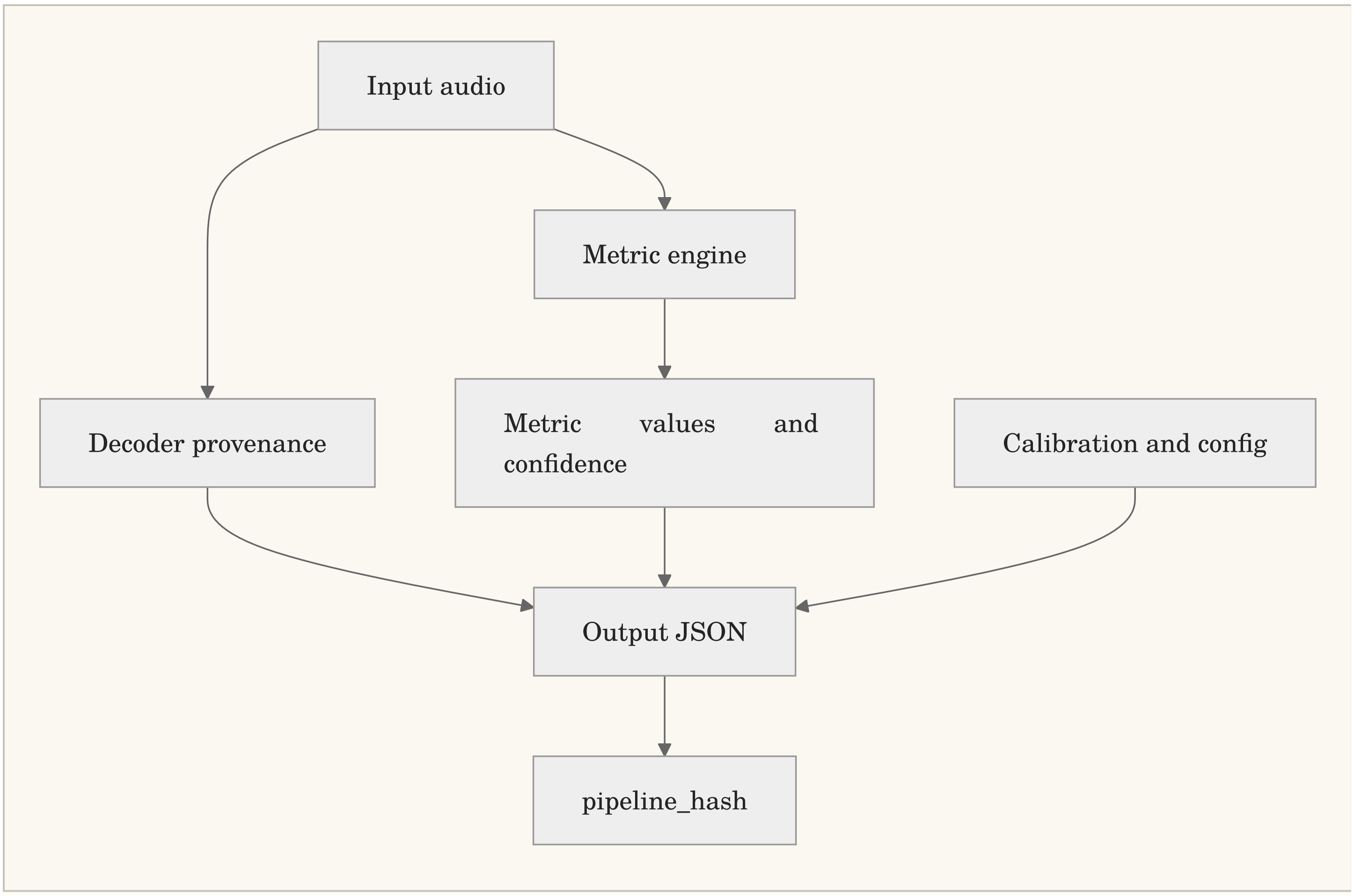
```
esl analyze input.wav --out-dir out --plot --similarity-matrix --novelty-matrix
```

For exact terminology and defaults, see Novelty and Anomaly and Similarity Search.

4. Read the Results Without Guessing

The analysis JSON contains four ideas:

- `metadata`: the file, decoder, channel layout, calibration, device, and assumptions
- `metrics`: values, time series, units, confidence, and metric definitions
- `artifacts`: paths to larger sidecars such as FrameTables
- `pipeline_hash`: a compact fingerprint of the resolved configuration, metrics, and runtime library versions



`pipeline_hash` is built deterministically from the resolved configuration, metric list, window/hop settings, and library versions:

> **Equation preview.** At a high level, this is a provenance fingerprint: it combines the processing choices that must match before two analyses can be called equivalent.

$$H = \mathrm{hash}(C, M, W, L)$$

where C is the resolved configuration, M is the selected metric list, W is the frame/hop configuration, and L is the library-version map.

Plain English: if the hash changes, something important about the processing recipe changed. “I think we used the same settings” is not a reproducibility method.

Units matter

- `dBFS` : level relative to digital full scale; this needs no physical calibration.
- `SPL` , `dBA` , `dBC` : physical or weighted acoustic levels; these need an explicit calibration assumption.
- `ratio` , `Hz` , `s` , `ms` , `deg` : feature units, not sound-pressure levels.

Do not compare `dBFS` from two unrelated recorders as though it were a field measurement. Calibrate first.

5. Calibration: From Digital Samples to Useful Level Data

The minimum calibration mapping is simple:

```
dbfs_reference: 0.0
spl_reference_db: 94.0
weighting: A
```

Then run:

```
esl analyze recorder.wav --calibration calibration.yaml --out-dir out --json out/ana
```

For a pressure-chain conversion, add microphone sensitivity, preamp gain, and ADC full-scale voltage. Validate the setup with a known tone:

```
esl calibrate check calibration_tone.wav --calibration calibration.yaml --out out/ca
esl calibrate verify
```

The basic mapping is:

> **Equation preview.** At a high level, this expression makes the preceding method reproducible by stating the quantities and relationships used to compute the result.

$$L_{\mathrm{SPL}} = L_{\mathrm{dBFS}} + (L_{\mathrm{ref}} - L_{\mathrm{dBFS,ref}})$$

where L_{ref} is the known physical reference level and $L_{\mathrm{dBFS,ref}}$ is its measured digital level.

Plain English: calibration establishes the offset between what the recorder stores and what the microphone experienced. Keep the profile with the project.

Read Schema and Metrics Reference before claiming a calibrated result in a report or publication.

6. Multi-Channel, Ambisonics, and Spatial Metadata

`esl` preserves channels rather than silently forcing a downmix. Outputs state how aggregate values were made, and spatial metrics declare their assumptions.

For an Ambisonics or array recording whose filename is not enough to identify the convention, use a sidecar:

```
{
  "layout_family": "ambisonic",
  "layout_hint": "ambisonic_b_format",
  "channel_labels": ["W", "Y", "Z", "X"],
  "ambisonics": {
    "order": 1,
    "component_order": "ACN",
    "normalization": "SN3D"
  }
}
```

```
esl analyze field_array.wav --spatial-metadata-sidecar field_array.spatial.json
```

The sidecar can clarify channel labels, Ambisonics order, channel order, normalization, layout, and geometry. It cannot change the number of channels the decoder actually found. That is intentional.

For terminology and full sidecar behavior, see Shard Workflows.

7. Architectural and Impulse-Response Work

For an impulse response, request the architectural metrics explicitly:

```
esl analyze room_ir.wav --out-dir out --json out/room_ir.json \
  --metrics rt60_s,edt_s,clarity_c50_db,clarity_c80_db,definition_d50
```

Typical questions:

- How long does sound persist? Use `rt60_s` and `edt_s`.
- Is speech likely to retain early energy? Use `clarity_c50_db` and `definition_d50`.
- Is music detail likely to remain distinct? Use `clarity_c80_db`.

The decay model is commonly summarized as:

> **Equation preview.** At a high level, this is a model rather than an exact identity: it summarizes an observed process with stated assumptions and an expected approximation error.

$$E(t) \approx E_0 10^{-3t/T_{60}}$$

where $E(t)$ is decay energy at time t, E_0 is initial energy, and T_{60} is the 60 dB decay time.

Plain English: the metric estimates how quickly a room response fades. Inspect the fit quality and validity flags, especially if the tail has weak SNR.

See Metrics Reference for definitions and limitations.

8. Long Files, RF64, and Shards

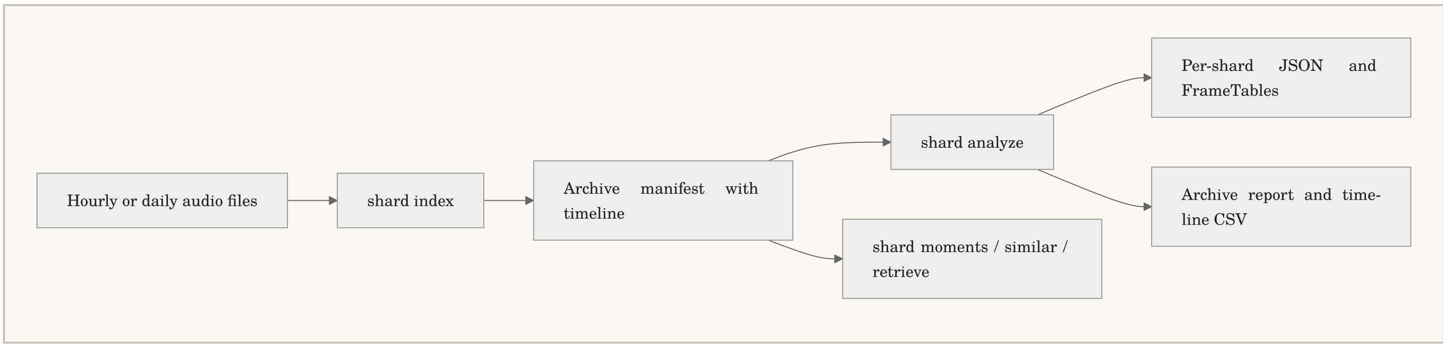
Do not load a multi-day or multi-year recording just because the filename ends in `.wav`. Use chunking for one large file:

```
esl analyze daylong.rf64 --out-dir out \
  --chunk-hours 1 --streamable-only --summary-only \
  --frame-table-csv out/frame_table.csv \
  --checkpoint-dir out/checkpoints --resume
```

If the deployment already produces hourly or daily files, make a shard manifest:

```
esl shard index archive/ --out out/archive_manifest.json \
  --calendar-start 2026-01-01T00:00:00Z \
  --calendar-timezone America/New_York

esl shard analyze out/archive_manifest.json --out out/archive_analysis \
  --chunk-hours 1 --streamable-only --summary-only \
  --frame-table-dir out/frame_tables --resume
```



Use `esl shard moments` to find the most novel events across the full archive, and `esl shard retrieve` to find query-like time windows. For a representative performance check before a huge spatial query, run:

```
esl shard profile-retrieve out/archive_manifest.json query.wav \
  --out out/spatial_profile --max-shards 100 --spatial-mode append
```

Read [RF64 and Large Files] and [Shard Workflows] before planning a multi-year run.

9. FrameTables and ML Exports

Use `--ml-export` for a single analysis product, or FrameTable sidecars for long runs:

```
esl analyze input.wav --out-dir out --ml-export

esl shard dataset out/archive_analysis/shard_analysis_report.json \
  --out out/archive_dataset_manifest.json --split-ratios 0.8,0.1,0.1
```

The canonical tensor layout is:

> **Equation preview.** At a high level, this defines a set or eligibility condition: it states which items qualify for a later analysis or decision step.

$$\mathbf{T} \in \mathbb{R}^{C \times F \times K}$$

where C is channel count, F is frame count, and K is the number of feature columns.

Plain English: tabular exports give classical ML one row per frame; tensor exports give deep-learning systems a declared channel, frame, feature layout. Do not infer array axes from whatever shape happened to arrive in NumPy.

See ML Features for naming rules, split semantics, and large archive guidance.

10. Input Formats, Decoders, and Preparing Audio

What files can `esl` open?

Native decoding through SoundFile covers WAV, RF64, FLAC, AIFF/AIF, and CAF. When a format needs it, `esl` uses FFmpeg/FFprobe for MP3, AAC, OGG, Opus, WMA, ALAC, M4A, and related container variants. SOFA impulse responses are supported when the required HDF5 dependency is installed.

You have	Recommended action	Important note
ordinary WAV or FLAC	run `esl analyze` directly	SoundFile is normally used
MP3, AAC, M4A, or Opus	run `esl doctor input.ext` first	install FFmpeg if decoding fails
a WAV approaching 4 GB	use RF64 for future captures	classic RIFF WAV has a size limit
many hours in separate files	make a shard manifest	do not concatenate just to analyze
SOFA spatial IR	run analysis with the SOFA file	inspect channel and spatial metadata

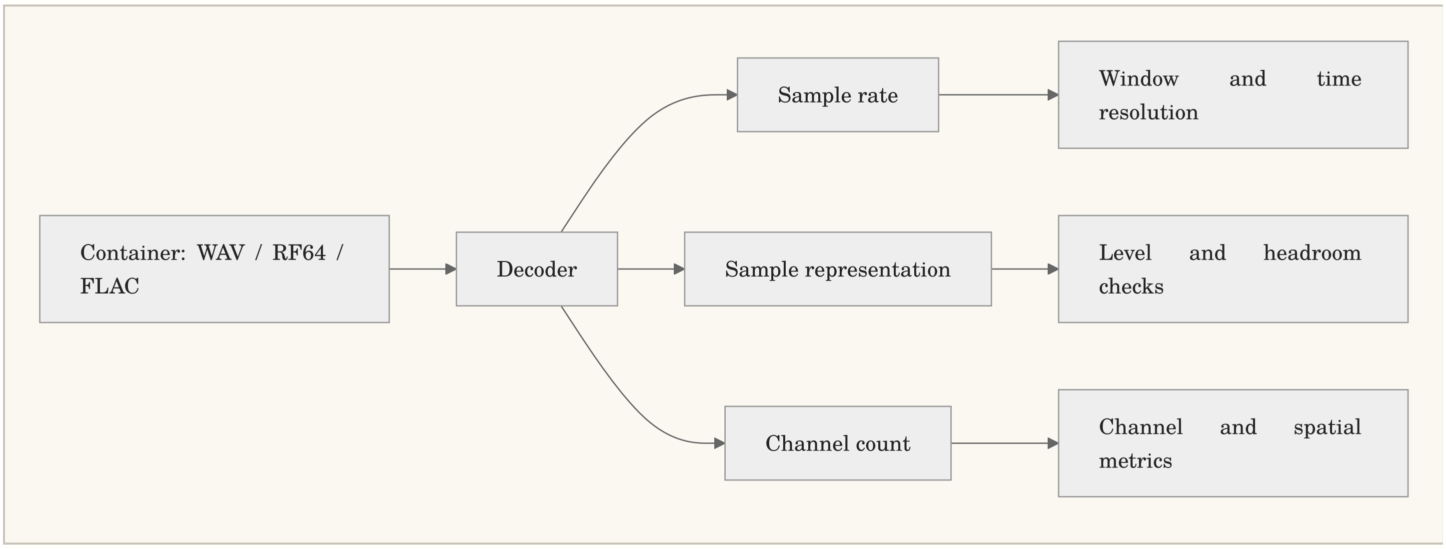
Use this before a difficult first run:

```
esl doctor input.wav
```

The doctor output tells you the available decoder path, sample rate, channels, duration, file size, and common long-file risks. Its purpose is not to judge your audio. It is to stop a 12-hour analysis from failing 11 hours and 58 minutes in because FFmpeg was missing.

Sample rate, bit depth, and channels

Sample rate tells you the time grid; bit depth or floating point encoding tells you how samples are represented. Channel count tells you how many simultaneous signals exist. They are different facts, and confusing them makes troubleshooting unnecessarily theatrical.



`esl` stores decoder provenance in output JSON. If a compressed file passes through FFmpeg, the result records the FFmpeg version and FFprobe stream summary. That is useful when two machines disagree about an awkward legacy file.

Resampling deliberately

Use `--sample-rate` only when you want a target processing rate:

```
esl analyze input_96k.wav --sample-rate 48000 --out-dir out --plot
```

Resampling can save storage and processing time, but it removes information above the new Nyquist frequency. If ultrasonic content, high-frequency insect calls, or source localization cues matter, select the rate as a scientific decision, not a convenience flag.

11. Metrics: Ask a Small Question First

The metric catalog is intentionally broad. Start with the question, then choose the metric family instead of collecting every number because storage is cheap.

Question	Useful first metrics	What they describe
Is the file loud or clipped?	`rms_dbfs`, `peak_dbfs`, `crest_factor`, `dc_offset`	digital level, headroom, and signal-health clues
Is there useful signal above background?	`snr_db`, `noise_floor_dbfs`	a level contrast estimate and confidence
What frequencies dominate?	spectral centroid, bandwidth, rolloff, flatness	spectral balance and texture
Is something changing?	`novelty_curve`, spectral flux, change detection	frame-to-frame difference
Is an ecosystem soundscape diverse?	NDSI and ecoacoustic indices	band-energy balance and temporal structure
Does a room response decay well?	RT60, EDT, C50, C80, D50	reflection and decay behavior
Does a multichannel scene differ spatially?	IACC, ILD, ITD, coherence	channel relationship proxies

For the authoritative definition, formula, units, aggregation rule, calibration dependency, and streamability of every metric, use Metrics Reference.

A reliable level example

For a sample sequence $x[n]$ with N samples, RMS is:

> **Equation preview.** At a high level, this is a normalized relationship: it compares a selected quantity with an appropriate reference, total, or scale factor.

$$x_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x[n]^2}$$

where $x[n]$ is the sample at index n and N is the number of samples in the measurement window.

The corresponding dBFS value is:

> **Equation preview.** At a high level, this is a logarithmic scale conversion: it compresses a wide range of linear values into decibels or another interpretable log scale.

$$L_{\mathrm{RMS,dBFS}} = 20\log_{10}\left(\max\left(x_{\mathrm{RMS}}, \epsilon\right)\right)$$

where ϵ is a very small positive floor that keeps silence finite.

Plain English: RMS is a stable estimate of average signal strength. Peak tells you the tallest instant. Crest factor compares the two. None of them becomes SPL until you supply a physical calibration relationship.

Confidence and validity flags are not decoration

Some metrics can be uncertain because the signal is too quiet, the tail is too short, a fit is poor, or a required calibration assumption is missing. Check:

- `metadata.validity_flags`
- each metric's `confidence`
- warnings in `metadata.warnings`
- IR fit-quality details for room metrics

If a metric has a low confidence, do not discard it automatically. Use it as a signal to inspect the waveform, spectrogram, SNR, and assumptions before making a claim.

12. Windows, Hops, and Time Resolution

Most time-varying features are calculated on overlapping frames. `--frame-size` controls how much audio each calculation sees; `--hop-size` controls how often the calculation moves forward.

```
esl analyze input.wav --frame-size 4096 --hop-size 1024 --out-dir out
```

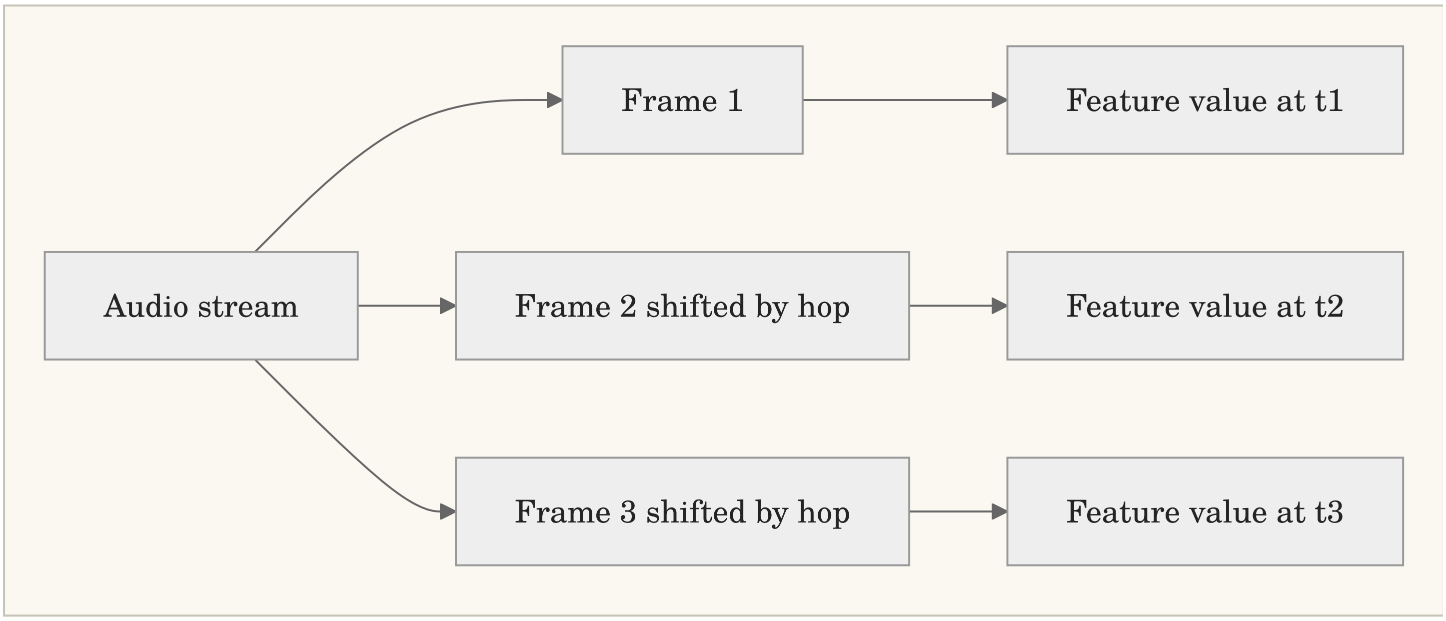
At sample rate f_s, frame duration and hop duration are:

> **Equation preview.** At a high level, this is a normalized relationship: it compares a selected quantity with an appropriate reference, total, or scale factor.

$$T_{\text{frame}} = \frac{N_{\text{frame}}}{f_s}, \qquad T_{\text{hop}} = \frac{N_{\text{hop}}}{f_s}$$

where N_{frame} and N_{hop} are frame and hop sizes in samples, and f_s is sample rate in Hz.

Plain English: a longer frame gives more frequency detail but blurs rapid events; a shorter frame tracks quicker changes but gives coarser frequency detail.



Practical starting points:

Material	Frame / hop starting point	Why
speech, general soundscape	2048 / 512 samples	balanced default
transient-rich events	1024 / 256 samples	quicker time response
slow level trends	4096 / 1024 samples	smoother summaries
very long archive reporting	chunks in minutes or hours, frames in samples	memory-safe processing with normal feature resolution

For minute-, hour-, or day-scale analysis chunks, use `--chunk-minutes`, `--chunk-hours`, or `--chunk-days`. A chunk is a memory-management unit; it is not necessarily the same thing as a feature frame. See Signal and Window Visual Guide.

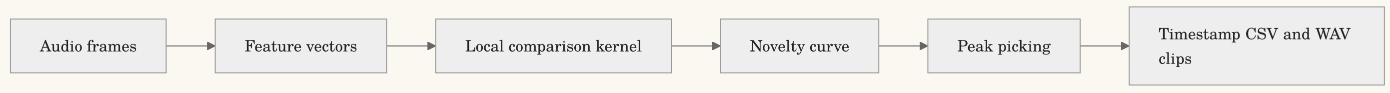
13. Novelty, Anomaly, and Interesting Moments

Novelty is a measure of change relative to nearby audio. It is not a universal definition of importance. A quiet rare bird call can be important and have a modest novelty score; a slammed door can be very novel and scientifically boring. The right workflow combines ranking with listening and context.

The standard first pass

```
esl analyze input.wav --out-dir out --plot --novelty-matrix
esl moments extract input.wav --out out/moments --top-k 10 --rank-metric novelty_cur
```

The novelty curve begins with a feature representation, compares nearby frames, rectifies or normalizes the result, and can use peak picking to nominate events.



A simple spectral-change form is:

> **Equation preview.** At a high level, this is an accumulation: it combines contributions over samples, channels, bins, or frames into one quantity.

$$v[t] = \sum_k \max\left(0, X[k,t] - X[k,t-1]\right)$$

where $X[k,t]$ is a nonnegative feature value at frequency or feature index k and frame t.

Plain English: this counts positive upward changes from one frame to the next. Different novelty methods use richer features and kernels, but the intuition is the same: find moments that differ from their local neighborhood.

Similarity matrix versus novelty matrix

- A similarity matrix compares every selected frame to every other selected frame. Repeated scenes become blocks or diagonal patterns.
- A novelty matrix applies a local contrast operation to self-similarity, emphasizing boundaries between sections or acoustic states.

Use a similarity matrix when you ask “when does this sound recur?” Use a novelty matrix when you ask “where does the scene change?” Read Novelty and Anomaly for kernel, normalization, and peak-picking defaults.

Do not over-interpret an anomaly score

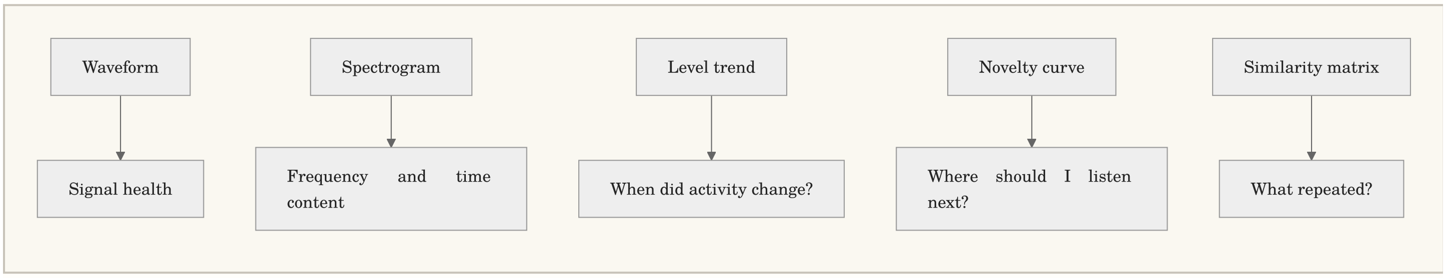
An anomaly or isolation-forest score is a ranking signal, not a species label, incident report, or causal explanation. Export the scores, inspect the selected clips, and preserve the configuration hash so another analyst can reproduce the candidate list.

14. Plots That Help You Decide What to Do Next

Use `--plot` for static PNGs. Use `--interactive` when a Plotly output helps you inspect a long time series. Add `--show` only when you want the operating system to open generated files automatically.

```
esl analyze input.wav --out-dir out --plot --interactive \
  --plot-metrics rms_dbfs,spl_a_db,novelty_curve,snr_db
```

Plot	Good first question	What to look for
waveform	is there clipping or silence?	flat tops, DC drift, recording gaps
spectrogram	what occupies the spectrum over time?	bands, impulses, periodic calls, machinery
mel spectrogram	what changes perceptually?	broad energy patterns and repeated texture
LTSA	what dominates over a long period?	persistent tonal or broadband energy
SPL/dBFS over time	when did level change?	activity cycles and obvious artifacts
SNR over time	when is analysis trustworthy?	low-confidence segments
novelty curve	where should I listen?	isolated peaks and sustained transitions
RT60 decay	is an IR fit plausible?	usable decay range and fit quality



When a graph surprises you, export the surrounding audio rather than adjusting parameters until the graph agrees with your expectation. The audio is allowed to have the last word.

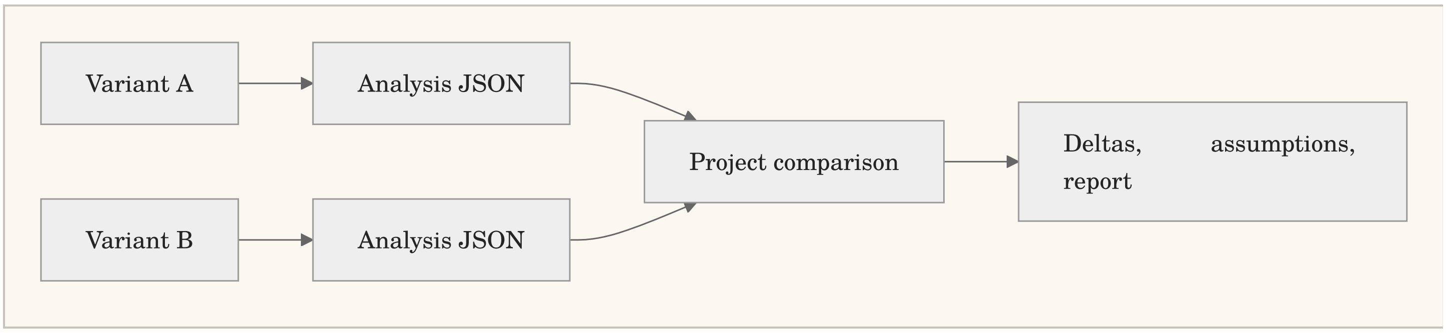
15. Batch Analysis, Projects, and Design Variants

Use `batch` for independent files such as a corpus. Use `shard analyze` for ordered pieces of one deployment. This distinction matters because an archive has a timeline and a corpus does not.

```
esl batch recordings/ --out out/batch --csv --parquet --hdf5 --plot
```

For architectural design variants or a named study, retain context with project and variant values:

```
esl analyze restaurant_design_A.wav --project restaurant_design --variant A --out-di
esl analyze restaurant_design_B.wav --project restaurant_design --variant B --out-di
esl project compare --project restaurant_design --root out --baseline A
```



Keep every variant's source audio, calibration profile, metrics, and window settings aligned. A difference between A and B is only useful if the workflow behind the difference stayed comparable.

16. Interoperability and Export Choices

`esl` can emit JSON, CSV, Parquet, HDF5, MATLAB `.mat`, and compatibility-style CSV outputs for measurement workflows. Choose the file type based on how it will be used next.

Format	Choose it when	Avoid it when
JSON	provenance, nested metadata, a single analysis report	you need millions of frame rows
CSV	a person needs to inspect or import a small table	data types or scale matter
Parquet	analytics, dataframes, long FrameTables	the recipient only has a spreadsheet
HDF5	appendable numeric arrays and scientific tooling	a plain-text review is needed
MAT	MATLAB workflows	a language-neutral archive is required

Example:

```
esl analyze input.wav --out-dir out \
  --json out/analysis.json --csv out/summary.csv \
  --parquet out/summary.parquet --hdf5 out/analysis.h5 --mat out/analysis.mat
```

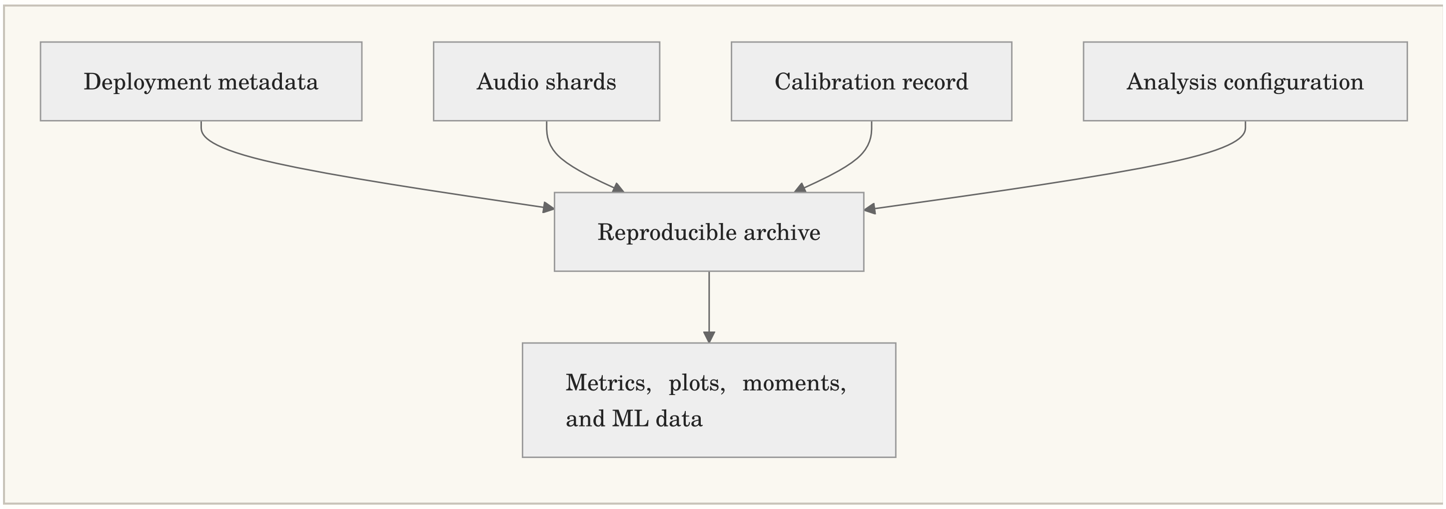
The output schema records weighting curves, level definitions, configuration, and provenance. Compatibility CSV is an exchange starting point, not a claim that every vendor application has identical hidden assumptions. Document your window duration, weighting, calibration mapping, and channel aggregation when cross-checking with another system.

17. Environmental Acoustic Monitoring Workflows

Environmental audio projects are strongest when they treat recordings as measurements with context, not just files with evocative names. Keep a small deployment record beside the audio that states:

- recorder model, microphone, gain, and channel mapping
- site identifier and coordinates at the precision appropriate for the project

- deployment start/end time and time zone
- sample rate, bit depth, and recording schedule
- calibration method and reference tone details
- weather, maintenance, and known disturbances



A practical field workflow

```
# 1. Index an hourly or daily deployment with a real archive clock.
esl shard index deployment_audio/ --out out/manifest.json \
  --calendar-start 2026-04-01T00:00:00Z --calendar-timezone America/New_York

# 2. Run a streaming-safe archive pass with FrameTable sidecars.
esl shard analyze out/manifest.json --out out/analysis \
  --chunk-hours 1 --streamable-only --summary-only \
  --frame-table-parquet-dir out/frame_tables --checkpoint-dir out/checkpoints --resu

# 3. Find candidate moments for listening and annotation.
esl shard moments out/manifest.json --out out/moments --top-k 50 \
  --rank-metric novelty_curve --window-before 10 --window-after 20

# 4. Summarize archive-level change and calmness proxies.
esl shard insights scene out/analysis/shard_analysis_report.json --out out/scene
esl shard insights calmness out/analysis/shard_analysis_report.json --out out/calmne
```

Use ecoacoustic indices as *summaries to compare with context*, not as a substitute for listening or annotation. An index can reveal a seasonal shift, equipment fault, human disturbance, or microphone wind artifact. It cannot decide which of those stories is true by itself.

Diversity, calmness, and soundscape change

`esl` provides soundscape-oriented features and archive insights that can help organize large deployments. Their most defensible use is comparative:

- compare a site to itself across time with the same capture chain
- compare treatment and control only after aligning calibration and schedule
- use visual trends to choose intervals for listening or annotation
- report the metric version, time scale, and exclusions

For a categorical frame-energy distribution p_i, entropy-style diversity has the familiar form:

> **Equation preview.** At a high level, this is a logarithmic scale conversion: it compresses a wide range of linear values into decibels or another interpretable log scale.

$$H = -\sum_i p_i \log(p_i)$$

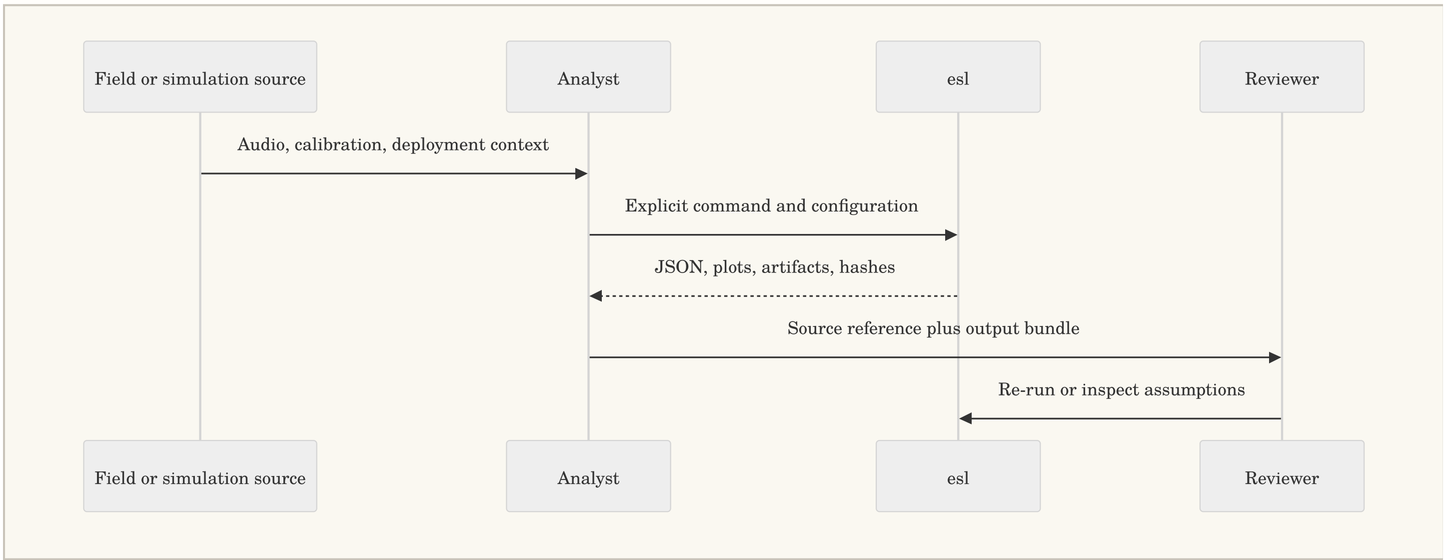
where p_i is the normalized share of energy or observations in category i.

Plain English: diversity measures how evenly a representation is spread. It does not mean "ecosystem health" unless the study validates that relationship in its own setting. See Metrics Reference and References.

18. Reproducibility, Review, and Research Handoff

Every serious analysis should be reviewable by someone who was not present when you ran it, including future you. `esl` writes the materials needed for that handoff directly into its outputs.

Keep	Why it matters
original audio or immutable archive reference	lets others verify the source
analysis JSON	preserves metrics, metadata, warnings, and confidence
calibration profile	explains physical-level mapping
configuration and pipeline hash	identifies the exact processing recipe
FrameTable sidecars	supports later statistics and ML without rerunning audio
moments CSV and clips	makes candidate event review auditable
README or lab notebook	explains the study question and exclusions



Deterministic defaults are necessary, not sufficient

Fixed seeds and configuration hashes prevent accidental drift. They do not repair an unclear question, a bad microphone placement, or a train/test split that lets near-identical time periods leak into both sides. Good reproducibility combines deterministic software with an explicit study design.

For ML work, consider holding out sites, devices, or time blocks rather than randomly shuffling adjacent frames. The correct split follows the claim you want to make. A model tested on another minute of the same recorder day is answering a much easier question than a model tested on another season or site.

19. Common Workflows

Goal	Start here
"What is in this file?"	`esl simple input.wav`
"Make a full report and plots."	`esl analyze input.wav --out-dir out --plot`
"Find the weirdest bit."	`esl moments extract input.wav --out out/moments --single`
"Find 33 unusual archive events."	`esl shard moments manifest.json --out out/moments --top-k 33`
"Find recordings like this one."	`esl similar query.wav corpus/ --out out/similar`
"Find matching moments inside an archive."	`esl shard retrieve manifest.json query.wav --out out/retrieve`
"Export ML-ready data."	`esl analyze input.wav --out-dir out --ml-export`
"Work safely on a huge file."	`esl analyze input.rf64 --chunk-hours 1 --streamable-only --summary-only`

More examples live in Task Recipes and the easy-script catalog.

20. When Something Goes Wrong

Start with the smallest diagnostic that can tell you something useful:

```
esl doctor input.wav
esl schema
esl --help
```

Common fixes:

- Compressed formats fail: install FFmpeg and make sure `ffprobe` is on `PATH`.

- `esl` is not found: activate your environment or run `python -m esl`.
- Huge file work fails: use RF64 where appropriate, chunking, FrameTable sidecars, checkpoints, and shards.
- SPL looks suspicious: check the calibration reference and verify a known tone.
- Spatial result looks wrong: supply a sidecar; do not rely on a filename guess.
- ML output is enormous: keep frame data as appendable CSV/Parquet/HDF5 and batch it downstream.

The full Troubleshooting Guide is deliberately more detailed. Audio has many ways to be politely uncooperative.

21. Where to Go Next

- Getting Started: first commands and common failure modes
- Task Recipes: copy-paste workflows by goal
- Metrics Reference: formulas, units, and assumptions
- Novelty and Anomaly: novelty curves and matrices
- Moments Extraction: event selection and clips
- Similarity Search: file and feature-vector comparison
- Shard Workflows: long archives and calendar timelines
- ML Features: FrameTable and tensor contracts
- References: papers, standards, and implementation context

The detailed documentation is there when you need it. The first command is still just:

```
esl analyze input.wav --out-dir out --plot
```

Appendix: One-Page Command Card

Use this as the "I know what I want, just remind me of the command" page.

Need	Command
Check installation or a file	`esl doctor [input.wav]`
Get a short summary	`esl simple input.wav`
Analyze and plot	`esl analyze input.wav --out-dir out --plot`
Extract one novel event	`esl moments extract input.wav --out out/moments --single`
Extract several events	`esl moments extract input.wav --out out/moments --top-k 20`
Compare a query to a folder	`esl similar query.wav corpus/ --out out/similar`
Index an archive folder	`esl shard index archive/ --out out/manifest.json`
Analyze a shard archive	`esl shard analyze out/manifest.json --out out/archive_analysis`
Find archive moments	`esl shard moments out/manifest.json --out out/moments --top-k 33`
Export ML features	`esl analyze input.wav --out-dir out --ml-export`
Check calibration	`esl calibrate check tone.wav --calibration calibration.yaml --out out/check.json`
View the output schema	`esl schema`

If you want the exact flags, run `esl <command> --help`. If you want an explanation of a result, start with the corresponding linked chapter above.